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Stanford Medicine Endowed Professor of Pathology, Professor of Genetics, Professor of Biomedical Data Science, Professor, by courtesy, of Computer Science



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STANFORD UNIVERSITY SCHOOL OF MEDICINE
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RE: Letter of support for Patrick Walsh

Dear CS Honors Program Committee,

I am writing to endorse Patrick Walsh for admission to the CS Honors Program and am happy to serve as his faculty sponsor. During his time at Stanford, Patrick has demonstrated his technical proficiency and his research potential.

During the Biology Summer Undergraduate Research Program last summer, Patrick worked under Jonas Cremer and investigated the role of bacterial strains inhabiting the human gut in the development of colorectal cancer by analyzing their metabolic byproducts and computationally modeling in vivo conditions. This summer, I have had the opportunity to serve as his faculty mentor for the CS+Social Good internship. Over the past three months, he has worked on developing AI/ML open-source tools for infectious disease research in the global south.

Patrick's proposed thesis on creating a privacy-protecting matching system for rare disease phenotypes is innovative and aligns well with the focus of my lab. He is well-poised to apply the skills learned at Stanford and through his work experience to create a privacy preserving matching layer by prototyping immediately on synthetic Phenopackets and MMIC-IV, followed by evaluation on STARR-OMOP-deid as access is granted.

While pursuing his B.S. on the Computational Biology track, he has also excelled in his pre-med coursework and extracurricular activities. His technical background in CS and preliminary experience with healthcare have positioned him to produce an innovative and impactful project if he is accepted to the program. Given his interest in pursuing his MD and continuing his research career after graduation, I strongly recommend that Patrick be given the opportunity to pursue this project. I am confident that he has the skills and motivation to produce an excellent thesis and become a valued member of the CS Honors program community.

All the best,

Stephen B. Montgomery, Ph.D.
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